

CHAPTER 2

PROBABILITY

2.1 - Elements of probability -Keywords

1)Experiment: Any process of observations or measurements that has well defined set of outcomes.

- **Deterministic**

The experiment that we conduct to verify the laws of science or established laws of other areas.

Ex: An experiment conduct to verify the Newton's law of motion

- **Probabilistic (Random) Experiment**

Experiments whose result is uncertain are called probabilistic or random experiments.

There are more than one possible results or outcomes (hence the results of a random experiment would be one of the possible results)

Ex: Tossing a coin

Roll a die

Keywords

2)Outcomes :

The results obtained from an experiment.

They can be instrument readings (length, weight, speed), counts, “yes” or “No” answers, or values obtained through extensive calculations are called outcomes of the experiment.

Keywords

3) Sample space:

The set of all possible outcomes of an experiment is called the sample space. It is usually denoted by S or Ω . Each outcome of a sample space is called an experiment or a sample point.

- If a sample space has a finite number of elements we may list the elements in the usual set notation.

Example: sample space for the possible outcomes of one flip of a coin.

Keywords

Sample space with a large or infinite number of elements is best described by a statement or rule.

Example: If the possible outcomes of an experiment are patient having leukemia, the sample space may be written as

$$S = \{x | x \text{ is a Leukemia Patient}\}$$

This is read as S is the set of all x such that x is a patient having leukemia.

Similarly if S is a set of odd positive integers

How we formulate sample space for a given situation will depend on the problem at hand.

If an experiment consists with one roll of a die and we are interested in which face is turned up, we would use the sample space.

However, if we are interested only in whether the face turned up is even or odd. Then we would use sample space.

Example: Describe a sample space that might be appropriate for an experiment in which we roll a pair of dies one red and one green.

Where x represent the number turned up by the green die and y represents the number turned up by the red die.

Or else a second sample space can be defined as the total of the two numbers turned up by the two dies.

Sample space can be classified as

Finite – if it contains finite number of elements S_1 and S_2 are finite sample space

If a coin is flipped until a head appears, then we will get a sample space .

This is an infinite sequence (unending) of elements. But number of elements in the sample space can be identified distinctly. Hence it is said to be countable sample space.

If a sample space contains finite number of elements but the elements are countable, then it is said to be discrete.

If outcomes of an experiment are neither finite nor countable infinite it is said to be continuous sample spaces arises in practice whenever the outcomes of experiments are measurements of physical properties such as *temperature, speed, length* and etc. that are measured on continuous state.

4) Events:

Any event can be identified with a collection of points, which constitute a subset of an appropriate sample space is known as an event.

Events are represented by capital letters.

Example: With reference to the sample space obtained by considering one roll of a die, an event A is defined as the number of points rolled with the die is divisible by 3.

Solution:

Sample space $S=\{1,2,3,4,5,6\}$

Only 3 and 6 are divisible by 3, then

$A=\{3,6\}$

Example: With reference to the example of rolling a pair of dice, green and red, define the events A, B and C as follows:

A=outcomes where the sum is 7

B= outcomes where the sum is less than or equal to 5

C=outcomes where green dice has higher value than red dice

Example: Consider an experiment that tossing 3 coins simultaneously with the intention of observing the faces that tossed up.

Identify elements of following events.

$A = \{\text{Getting two tails from the experiment}\}$

$B = \{\text{Getting three tails from the experiment}\}$

$C = \{\text{Getting at least 2 tails}\}$

$S =$

$A = \{\text{Getting two tails from the experiment}\}$

$B = \{\text{Getting three tails from the experiment}\}$

$C = \{\text{Getting at least 2 tails}\}$

Example: Construct a sample space for the lifetime of a CFL bulb and indicate the event F that the bulb fails before the end of 12 month.

If t is the lifetime of the bulb in months,

In many problems of probability we are interested in the events that are actually combinations of two or more events formed by taking Unions, Intersections and Complements.

If A and B are any two subsets of a sample space S,

Their Union

$$A \cup B$$

: Subset of S that contains all the elements that are either in A, in B or in both.

Their Intersection

$$A \cap B$$

: Subset of S that contains all the elements that are in both A and B.

the Complement A' of A

: Subset of S that contains all the elements of S that are **not** in A .

Two events A and B in sample space S are said to be mutually exclusive (ME) if there are no elements in common.

$A \cap B = \phi$, ϕ denotes the empty set which has no elements at all.

A contains in B

$$A \subset B$$

Example:

Suppose for a random experiment, the Sample Space, Events A,B and C are defined as follows:

$$S=\{A,B,C,D,E,F,G\} \quad A=\{A,B,C,D\} \quad B=\{A,C,E,G\} \quad C=\{E,F,G\}$$

Find the followings:

a) $A' =$

$B' =$

$C' =$

b) $A \cup B =$

c) $A \cap C =$

d) $B \cup C =$

e) $A \cap B' =$

f) $(A \cup B)' =$

g) $A' \cap B' =$

$$S=\{A,B,C,D,E,F,G\} \quad A=\{A,B,C,D\} \quad B=\{A,C,E,G\} \quad C=\{E,F,G\}$$

Find the followings:

$$\text{a) } A' = \{E,F,G\} \qquad B' = \{B,D,F\} \qquad C' = \{A,B,C,D\}$$

$$\text{b) } A \cup B = \{A,B,C,D,E,G\} \qquad \text{c) } A \cap C = \{\}$$

$$\text{d) } B \cup C = \{A,C,E,F,G\}$$

$$\text{e) } A \cap B' = \{B,D\}$$

$$\text{f) } (A \cup B)' = \{F\}$$

$$\text{g) } A' \cap B' = \{F\}$$

2.2 Axioms of Probability

1) Empirical probability (Frequency Interpretation)

- Results are completely based on experimental outcomes and no assumptions made.
- Probability of an event (A) is represented as $P(A)$

Ex: Perform an experiment n times and suppose the experimenter is interested about an event A .

One trial of the experiment would result either A or A' .

Out of n trials suppose A occurs m times. ($0 \leq m \leq n$)

The ratio m/n is proportion of times A occurs.

When n gets larger the ratio $\frac{m}{n}$ approaches to a constant value.

This ratio: m/n would be an estimate of $P(A)$.

Probability is defined based on law of Large Numbers.

[Law of Large Numbers | Statistical Convergence](#)

2. Subjective Probability

These probabilities express the strength of one's belief with regard to the uncertainties. These expressions are based on one's experience and knowledge. There are no evidences to prove them.

3. Axiomatic Probability

Probabilities are defined as “Mathematical objects” that behave according to certain well-defined rules.

Axioms of Probability

Suppose that each event A of an experiment having a sample space S there is a number, denoted by $P(A)$, that is in accord with the following three axioms.

AXIOM 1

The probability of an event is a non-negative real number.

i.e. $P(A) \geq 0$ for any subset A of S .

Axioms of Probability

AXIOM 2

$$P(S) = 1$$

AXIOM 3

If $A_1, A_2, A_3, \dots$ is a finite or infinite sequence of mutually exclusive events of S (i.e. events for which $A_i \cap A_j = \phi$ when $i \neq j$)

$$p\left(\bigcup_{i=1}^n A_i\right) = P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots \quad n=1, 2, \dots, \infty$$

We call $P(A)$ the probability of the event A .

The above axioms do not need proof. But if the resulting theory is to be applied, we must show that the axioms are satisfied.

Ex: An experiment has four possible outcomes A, B, C, and D that are mutually exclusive. Why the following assignments of probabilities are not permissible.

$$(a) P(A) = 0.12, P(B) = 0.63, P(C) = 0.45, P(D) = -0.20$$

$$(b) P(A) = \frac{9}{120}, P(B) = \frac{45}{120}, P(C) = \frac{27}{120}, P(D) = \frac{46}{120}$$

Probability of an Event

This is a measure of likelihood that it will occur on a scale from **0 to 1**.

Probability of 1 implies that the event in equation will occur 100% of the time or in other words, that is certain to occur.

In actual practice probabilities are assigned on the basis of past experience, on the basis of careful analysis of all underlying conditions, on the basis of subjective judgments or on the basis of assumptions.

Theorem 2.1

If A is an event in a discrete sample space S , then $P(A)$ equals the sum of the probabilities of the individual outcome comprising A .

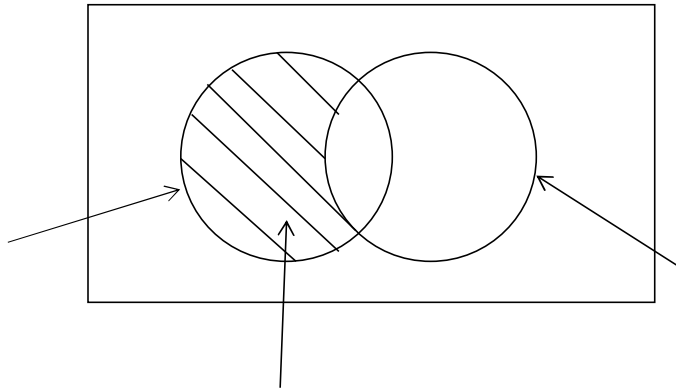
$$A = O_1 \cup O_2 \cup O_3 \dots$$

and since the individual outcomes O 's are mutually exclusive, from the third Axiom;

$$P(A) = P(O_1 \cup O_2 \cup O_3 \dots) = P(O_1) + P(O_2) + P(O_3) + \dots$$

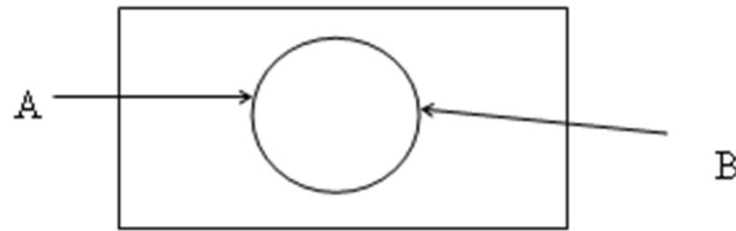
Basic Set Operations

$A-B$: A minus B is the set of all the elements from A that are not in B.



Basic Set Operations

Two sets A and B are equal if $A \subset B$ and $B \subset A$



Basic Set Operations

Distributive Law

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Basic Set Operations

De Morgan's Law

$$(A \cap B)' = A' \cup B'$$

$$(A \cup B)' = A' \cap B'$$

If A is contained in B $A \subset B$ then,

$$A \cap B = A \text{ and } A \cap B' = \phi$$

Example: If we twice flip a balanced coin, what is the probability of getting at least one head?

Sample space $S =$

Since we assume that the coin is balanced these outcomes are equally likely.

Hence probability of a sample point =

Let A is getting at least one head, $A =$

Using theorem 2.1,

$P(A) =$

Example: A die is loaded in such a way that each odd number is twice as likely to occur as each even number. Find $P(G)$ where G is the event that a number greater than 3 occurs on a single roll of the die.

Sample space $S =$

Let probability for each even number =

Then probability for each odd number =

$G =$

$P(G) =$